

ComMod : a Companion Modelling Approach

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Introduction

The present signatories of this charter have worked for a number of years in the field of renewable resource management, using various tools, particularly Agent-Based Models and Role-Playing Games, to tackle issues regarding decision processes, common property, co-ordination among actors, etc. Working with models and games has been a mean to cross disciplines boundaries and to acknowledge the complex nature of the systems under study. This choice led us to formalise our relation to modelling within a companion modelling approach (Bousquet, Barreteau et al. 1999). At a time when models and simulations to tackle complexity and for decision support are flourishing, we found important to precise the contents of this approach, which should be understood as a scientific posture more than a modelling handbook. Modelling is merely an intermediary object (Vinck 1999) facilitating our collective and interdisciplinary thought.

The team posture

Our research activities are involved in the development process, and addressing genuine stakes and implementing them in the field is our preferred way to test and to question theories. We are dealing with a combination of pragmatic and theoretical questions regarding the management of renewable resources and the environment, and are facing complex and very dynamic research objects. Such a context leads to pay importance to uncertainty, and to the existence of multiple and legitimate points of view, including the ones produced by scientific expertise. These different view points deserve to be taken into account in an iterative process of understanding, confrontation, and analysis. Therefore, we have chosen to give ourselves a rigorous and refutable¹ doctrine which could be evaluated. It means that:

- a. The fate of all the assumptions backing modelling work is to be discarded after each interaction with the field, that is to say to be voluntarily and directly subjected to refutation,
- b. Having no *a priori* implicit experimental hypothesis is an objective implying the adoption of procedures to unveil such implicit hypotheses,
- c. The impact in the field has to be taken into consideration as soon as the first steps of the approach, in terms of research objectives, quality of the approach, quantified monitoring and evaluation indicators.
- d. Particular attention should be given to the process of validation of such a research approach, knowing that a general theory of model validation does not exist, and that procedures differing from those used in the case of physical, biological, and mathematical models need to be considered.

¹ In research, refutation is the only rigorous expression of intellectual freedom, particularly in the ambiguous context of action-research.

The present first version of this document (“Charter 1.0”) results from a long discussion, and numerous common activities among its different signatories over the last two years. By definition, the present charter is an evolving one and will be periodically revised and updated with the agreement of its signatories. This is because struggling for rigor implies its improvement at each step, and striving for refutation leads to continuous questioning of our posture.

The common approach

In agreement with the complex and dynamic nature of the processes under study, our companion modelling approach requires a permanent and iterative confrontation between theories and field circumstances. Therefore, it is based on repetitive back and forth steps between the model and the field situation. Thus, this approach is adapted to the complexity and the openness of the systems under study because: i) it considers as legitimate and takes into account points of view which could possibly be contradictory, ii) it organises the compulsory questioning of any new element introduced in the approach, iii) during each loop, it confronts itself to new external elements.

Because in the field of cognitive sciences Multi-Agent Systems (MAS) are particularly adapted to the exploration of hypotheses presented as “true”, and to the representation of dynamic and complex systems (Janssen 2002), this simulation tool has been privileged in our approach and is used in association with other ones.

The importance given to field work in our approach leads to expectations regarding tangible effects at these sites. Depending on the experiences and tools used, the outputs could be of three kinds: the modification of perceptions, of behaviours, or of actions. Finally, one needs to distinguish between the use of this approach in two specific contexts: the production of knowledge on some complex systems, and the support to collective decision-making processes. While the first context corresponds to systems research *via* a particular relationship to field work, the second one corresponds to methodological research to facilitate a concerted management of such systems by proposing a particular relationship to field work to achieve such an end.

First objective: understanding complex environments

In this context, modelling deals with the dialectique among the researcher, the model and the field. Simulation accompanies an iterative research process, which is specific to each situation. The endless following cycle “field work-> modelling -> simulation -> field work again, etc.” corresponds to this concept. This leads to accept a diversity of models and methods, each contributing to a new kind of relationship between the simulation, the research itinerary, and the the decision-making process.

The researcher starts building a first preliminary model to explicit his/her theoretical as well as field-based pre-conceptions. The confrontation of this first model with real circumstances leads to revise and to re-build it, taking gradually into account the features of the field situation, but also the questions that stakeholders are asking to themselves. The discussion of the model hypotheses, and the simulations implemented according to an experimental plan corresponding to the initial questions, allows to modify the formers and to formulate new questions. This process leads to the construction of a new model, which is either derived from

the previous one following its confrontation with the real circumstances and its evolution, or an entirely new one. As this cycle repeats itself, we create a family of models representing the successive interactions between the researcher and the field. There is no gradual *a priori* complexification of a model incorporating more and more elements to fit with “the reality”.

Such a family of models is a genuine knowledge-based system allowing interacting researchers and stakeholders to increase their personal and common knowledge of the system, of the current processes, and of the situation of each actor-observer in such processes (Berkes and Folke 1998). Here, the key challenge of companion modelling is to deliver an improved understanding of these processes rather than a “turn key” itinerary for renewable resources management. As a consequence, there is a special relationship between the field and the model (*sensu lato*²). Instead of proposing a simplification of stakeholders knowledge, the model is seeking a mutual recognition of everyone representation of the problematique under study. Such mutual recognition lies on indicators which are gradually and collectively built during the implementation of the approach, and constitutes the fundamentals of participatory modelling.

The underlying hypothesis is that, in most of the renewable resource management situations, what actors need is less a simple formalisation of their own perception than an exchange among stakeholders (including experts) about such representations, and existing knowledge. By structuring these exchanges, the simulation helps the stakeholders to validate the interactions between different representations and the system dynamics integrated in the model. A true learning process on the system under study is taking place through interactions with and among local stakeholders (Conein and Jacopin 1994).

Second objective: to support collective decision-making processes in complex situations

In this case, the approach facilitates collective decision-making processes by making more explicit the various points of view and subjective criteria, to which the different stakeholders refer implicitly or even unconsciously. Indeed, as demonstrated in past research (Mermet 1992; Weber and Reveret 1993; Ostrom, Gardner et al. 1994; Funtowicz, Ravetz et al. 1999), when facing a complex situation, the decision-making process is evolving, iterative, and continuous. It means that this process produces always imperfect “decision acts”, but following each iteration they are less imperfect and more shared. In other words, the question is not the quality of the choice, but the quality of the process leading to it. It is not about finding the best solution, but to take into consideration as well as possible the uncertainties of the situation. To improve the quality of collective decision-making processes, the approach aims at elucidating and sharing the points of view determining them. This approach refers to a dynamic perception of the decision-making process in which the scientific and technical perception is only one among others, and not the pre-supposed right perception toward which the decision should be attracted. The objective is not to ambitiously produce decisions and definitive results, but to enrich the decision-making process in terms of technical (information, technical quality of actions launched, etc.), or sociological (greater concertation, reinforcement of stakeholders power in making decisions, etc.) aspects. Because we are

² Here, from verbal diagnosis to computer models, all possible types of dialogues between the researcher and the stakeholder are considered as, more or less formal, participatory modelling processes. As a matter of fact, in all cases, a representation is built, shared, and made formal in a particular way: “unconscious” modelling (purely verbal participatory diagnoses, expert cognitive representations of the situation, etc.), and more explicit models (maps, information systems, computer models, etc.).

dealing with an evolving, iterative, and continuous process, the way to accompany it should also bear the same characteristics.

What tools can participate in such a process, and how to use them to accompany the collective decision-making dynamics? That is to say how to help stakeholders govern a situation along a continuous and gradually enriched itinerary, instead of proposing ready-made expert solutions? This is a situation similar to experimental approaches in post-normal science in which, based on a shared conception of the evolution of the current situation, the stakeholders can be collectively engaged in a process to take uncertainties into account (Funtowicz and Ravetz 1994). We propose the use of various tools to accompany and to support the decision-making process: MAS, role-playing games, geographic information systems, economic tools, etc. Depending on the situation, the production of knowledge or points of view on a given system could lead to: i) an improved knowledge of actors/decision-makers, ii) a facilitated dialogue among stakeholders (including experts) providing a framework for discussion and sharing of information, an exchange of viewpoints, knowledge, and beliefs among them, iii) a negotiation support system aiming at closing the gap between diverging points of view in a given conflicting situation.

Here, even if it is not covering the whole process of mediation by itself, companion modelling is taking part in it. Stakeholders learn collectively by creating, modifying, and observing simulations. When carrying out simulations one acts on the decision-making process by creating or modifying representations. Companion modelling leads stakeholders to share representations and simulations taking into account possible decisions and actions related to their environment which are under consideration (management rules, new infrastructures, etc.). Meanwhile, companion modelling does not include the other possible steps of the mediation process dealing with a more quantified expertise (size of a new infrastructure, estimated production, etc.). Companion modelling intervenes upstream of the technical decision to support the reflexion of concerned actors, in order to produce a shared representation of the problematique, and to identify possible ways toward a process of collective management of the problem.

A joint use

We consider that the organization of action is a result emerging from a dynamic of interactions among individual and/or collective stakeholders. Such a dynamic is constrained by the understanding and perception that every actor has of others' actions, therefore of his own indicators concerning the environment he is sharing with others. Consequently, it is fundamental to distinguish rigorously between the two aboved-mentioned kinds of use of our approach, even if in practice they are often implemented simultaneously. The first kind of use looks for its scientific legitimacy in the production and relevance of knowledge, while the second one aims at improving the quality of collective decision-making processes.

In both cases, there is production of knowledge through the interaction among researchers and local stakeholders. But in the first situation, this production of knowledge (being for researchers, or for local actors through training activities) is the objective, while in the second case we make the hypothesis that it is a necessary element of the method to achieve the main objective of supporting collective decisions. This distinction is as much a methodological question than a epistemological and analytical one: nothing can guarantee that the tools and the methods tested in a given situation will be useful, efficient, and adapted in another one,

particularly regarding the posture of the researcher-modeller in the process. This is why we tackle these two modelling problematics differently. On the other hand, we think that it is necessary to consider them jointly, because the points of view produced by each of the two modelling situations are useful to elucidate the secondary effects created by one of them.

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If you are interested in this approach and would like to join the group of signatories of this charter to participate to their forum, please register at the Cormas web site (<http://cormas.cirad.fr/en/reseaux/ComMod/index.htm>)

References

- BERKES F and Folke C, (Eds.) (1998) *Linking ecological and social systems*. Cambridge: Cambridge University Press.
- BOUSQUET F, Barreteau O, Le Page C, Mullon C and Weber J (1999). "An environmental modelling approach. The use of multi-agent simulations", In Blasco F and Weill A (Eds.) *Advances in environmental and ecological modelling*: Elsevier.
- CONEIN B and Jacopin E (1994) Action située et cognition: le savoir en place. *Sociologie du Travail*, 4. pp. 475-500.
- FUNTOWICZ S, Ravetz J and O'Connor M (1999) Challenges in the use of science for sustainable development. *International Journal of Sustainable Development*, 1. pp. 99-108.
- FUNTOWICZ S O and Ravetz J R (1994) The worth of a songbird; ecological economics as a post normal science. *Ecological Economics*, 10. pp. 197-207.
- JANSSEN M, (Ed.) (2002) *Complexity and Ecosystem Management: The Theory and Practice of Multi-agent Approaches*. Edward Elgar Publishers.
- MERMET L (1992) *Stratégies pour la gestion de l'environnement, la nature comme jeu de société ?* Paris: L'Harmattan.
- OSTROM E, Gardner R and Walker J (1994) *Rules, games and common-pool resources*. The University of Michigan Press.
- VINCK D (1999) Les objets intermédiaires dans les réseaux de coopération scientifique. *Revue Française de Sociologie*, 40. pp. 385-414.
- WEBER J and Reveret J-P (1993) Biens communs : les leures de la privatisation. *Le Monde Diplomatique*, coll. *Savoirs*, 2. pp. 71-73.